Cloud Computing Multiple Choice Questions

Based on the provided document, here are the multiple-choice questions categorized by topic:

Chapter 1: Introduction to Cloud Computing

- 1. What is the NIST definition's model for cloud computing primarily about?
  - a) Building data centers
  - b) On-demand network access to shared configurable computing resources
  - c) Manufacturing hardware
  - d) Selling software licenses
- 2. Which is NOT one of the 5 essential characteristics of cloud computing (NIST)?
  - a) On-demand self-service
  - b) Rapid elasticity
  - c) Dedicated hardware
  - d) Measured service
- 3. In cluster computing, nodes are:
  - a) Heterogeneous and loosely coupled
  - b) Homogeneous and tightly coupled
  - c) Heterogeneous and tightly coupled
  - d) Homogeneous and loosely coupled
- 4. Grid computing uses which type of network?
  - a) High-speed LAN
  - b) Low-speed WAN
  - c) Bluetooth

    - d) Satellite only
- 5. According to Amdahl's Law, if N approaches infinity, the speedup is limited by:
    - a) $1/P$ (parallel fraction)
    - b) $1/S$ (serial fraction)
    - c) N (number of processors)
    - d) $P\times N$
- 6. Gustafson's Law differs from Amdahl's Law because it assumes:
    - a) Fixed problem size
    - b) No serial component
    - c) Problem size grows with number of processors
    - d) Only one processor is used
- 7. Which cloud deployment model is available to the general public?
    - a) Private Cloud
    - b) Community Cloud
    - c) Public Cloud
    - d) Hybrid Cloud
- 8. A composition of both public and private cloud is called:
    - a) Community Cloud
    - b) Multi-Cloud
    - c) Hybrid Cloud
    - d) Virtual Cloud

- 9. In IaaS, what does the user manage?
  - a) Only the application
  - b) VMs, OS, CPU, Storage, applications
  - c) Nothing
  - d) Only the network
- 10. Gmail and Office 365 are examples of:
  - a) IaaS
  - b) PaaS
  - c) SaaS
  - d) CaaS
- 11. Google App Engine and Heroku are examples of:
  - a) IaaS
  - b) PaaS
  - c) SaaS
  - d) DBaaS
- 12. Vector clocks are used to:
  - a) Synchronize system time
  - b) Detect causal relationships and concurrency between events
  - c) Encrypt messages
  - d) Compress data
- 13. Which technique addresses cloud latency by storing frequently accessed data closer to users?

    - a) Encryption
    - b) CDN (Content Delivery Network)
    - c) Load balancing
    - d) Virtualization
- 14. De-perimeterization in cloud computing refers to:
    - a) Adding more firewalls
    - b) Erosion of traditional network boundaries
    - c) Removing cloud services
    - d) Upgrading servers
- 15. The zero-trust security model assumes:
    - a) All internal traffic is safe
    - b) No entity is trusted by default, verify everything
    - c) Only external threats exist
    - d) Firewalls are unnecessary
- 16. In three-tier architecture, the three layers are:
    - a) Hardware, Software, Network
    - b) Presentation, Application, Data Access
    - c) Frontend, Middleware, Cache
    - d) Client, Server, Cloud
- 17. In SOA, the communication bus is called:
    - a) API Gateway
    - b) Message Queue

    - c) Enterprise Service Bus (ESB)
    - d) Load Balancer
- 18. Microservices architecture uses which data storage pattern?
    - a) Shared database for all services
    - b) Database per service
    - c) Single file storage
    - d) No database needed
- 19. SOA primarily uses which protocol?
    - a) REST
    - b) GraphQL
    - c) SOAP
    - d) WebSocket
- 20. The service contract in SOA is written in:
    - a) JSON
    - b) WSDL (Web Services Description Language)
    - c) YAML
    - d) XML Schema
- 21. Backend for Frontend (BFF) pattern creates:
    - a) A single backend for all clients
    - b) Separate backend services for different frontend types
    - c) No backend at all

    - d) A database per frontend
- 22. What computing model is popular for providing infrastructure on demand since 2005?
    - a) Personal Computing
    - b) Mobile Computing
    - c) Utility Computing
    - d) Quantum Computing
- 23. Serverless computing is characterized by:
    - a) User managing servers
    - b) Stateless, no infrastructure management, pay per execution
    - c) Dedicated hardware
    - d) Persistent connections
- 24. Edge computing processes data:
    - a) In a central data center
    - b) Closer to the data source
    - c) Only in the cloud
    - d) On the user's local machine only
- 25. SIMD stands for:
    - a) Single Instruction Multiple Data
    - b) Several Instructions Managed Data
    - c) Simple Input Multiple Devices
    - d) Secure Instruction Multi-Domain

- 26. Capital Expenditure (Capex) refers to:
    - a) Monthly electricity bills
    - b) Upfront costs for buying equipment
    - c) Employee salaries
    - d) Internet subscription fees
- 27. The cloud service model where providers manage everything and users just access via browser is:
    - a) IaaS
    - b) PaaS
    - c) SaaS
    - d) FaaS
- 28. Lamport timestamps are used for:
    - a) Data encryption
    - b) Ordering events in distributed systems
    - c) Load balancing
    - d) Memory management
- 29. Clock skew is caused by:
    - a) Software bugs
    - b) Different local clocks on nodes, hardware drift, network delays
    - c) Too many users
    - d) Excessive CPU usage
- 30. NTP stands for:

- a) Network Transfer Protocol
- b) Network Time Protocol
- c) New Technology Platform
- d) Node Transmission Procedure

Chapter 2: Virtualization

- 31. What is the primary advantage of para-virtualization over full-virtualization?
  - a) Better security
  - b) Improved performance through guest OS modification
  - c) Easier installation
  - d) Lower cost
- 32. Which technology enables running containers with near-native performance?
  - a) Full virtualization
  - b) Hardware emulation
  - c) OS-level virtualization
  - d) Binary translation
- 33. What happens when a guest OS tries to execute a privileged instruction in full virtualization?
  - a) The instruction executes directly on hardware
  - b) The system crashes
  - c) A trap occurs, and the hypervisor handles the instruction
  - d) The instruction is ignored
- 34. Which ring typically runs user applications in x86?

    - a) Ring 0
    - b) Ring 1
    - c) Ring 2
    - d) Ring 3
- 35. A hypervisor uses memory ballooning. The balloon inflates and consumes 2GB. What happens from the hypervisor's perspective?
    - a) Memory remains allocated to the guest VM
    - b) The hypervisor can reclaim and redistribute this physical memory to other VMs
    - c) Memory is permanently lost
    - d) Memory is compressed and stored on disk
- 36. How does MMU prevent DMA attacks in virtualized environments?
    - a) By encrypting all DMA transfers
    - b) By translating device-originated memory addresses through virtualized address spaces
    - c) By disabling DMA entirely
    - d) By monitoring network traffic
- 37. When a guest OS modifies its page table and the shadow page table becomes inconsistent, what technique maintains coherency?
    - a) Write-through caching
    - b) Page table write protection and trap-on-write
    - c) Immediate synchronization
    - d) Periodic batch updates
- 38. Type 1 hypervisor runs:

    - a) On top of a host operating system
    - b) Directly on bare metal hardware
    - c) Inside a container
    - d) In user space
- 39. VirtualBox is an example of:
    - a) Type 1 hypervisor
    - b) Type 2 hypervisor
    - c) Container runtime
    - d) Cloud provider
- 40. The Popek-Goldberg theorem states that for efficient VMM:
    - a) All instructions must be privileged
    - b) Sensitive instructions must be a subset of privileged instructions
    - c) All instructions must trap
    - d) No instructions should trap
- 41. In paravirtualization, privileged instructions are replaced with:
    - a) System calls
    - b) Binary translations
    - c) Hypercalls
    - d) Interrupts
- 42. KVM stands for:
    - a) Key Virtual Machine

    - b) Kernel-based Virtual Machine
    - c) Kubernetes Virtual Manager
    - d) Known Virtualization Method
- 43. In hardware-assisted virtualization, VMs run in:
    - a) Ring 0 only
    - b) Ring 3 only
    - c) VMX (non-root) mode
    - d) Ring 1
- 44. Shadow page tables map:
    - a) GPA to HPA
    - b) GVA directly to HPA
    - c) HPA to GVA
    - d) GVA to GPA
- 45. Extended Page Table (EPT) is pointed by:
    - a) CR3 register
    - b) EPTP (Extended Page Table Pointer)
    - c) Stack pointer
    - d) Instruction pointer
- 46. The TLB (Translation Lookaside Buffer) is a cache for:
    - a) CPU instructions
    - b) Address translations
    - c) File data

    - d) Network packets
- 47. During VM live migration, the service downtime occurs in which phase?
    - a) Push phase
    - b) Setup phase
    - c) Stop and copy phase
    - d) Pull phase
- 48. Pre-copy migration strategy:
    - a) Stops the VM first, then copies everything
    - b) Copies memory first, then stops and copies remaining
    - c) Only copies CPU context
    - d) Doesn't involve memory transfer
- 49. Memory sharing in virtualization works by:
    - a) Allocating more RAM
    - b) Sharing identical frames between VMs (scanning, hashing, mapping duplicates)
    - c) Compressing all memory
    - d) Using only disk storage
- 50. In x86, the CR3 register points to:
    - a) The stack
    - b) The page table
    - c) The interrupt descriptor table
    - d) The code segment

- 51. What is multiplexing in virtualization?
  - a) Creating one virtual object from multiple physical objects
  - b) Creating many virtual objects from one physical object
  - c) Emulating a different type of resource
  - d) Removing virtual objects
- 52. Context switching in CPU involves saving:
  - a) Only the program counter
  - b) The entire CPU context (PCB) in kernel stack
  - c) Just the cache
  - d) Only register values
- 53. The CPL (Current Privilege Level) is stored in:
  - a) Stack pointer
  - b) Code Segment (CS) register
  - c) CR3 register
  - d) Data register
- 54. Which is NOT a valid reason for VM live migration?
  - a) Load balancing
  - b) Hardware maintenance
  - c) Increasing network latency
  - d) Distributing workloads evenly
- 55. Segmentation divides memory into:
  - a) Fixed-size blocks

    - b) Logical segments (code, data, stack) of variable size
    - c) Equal partitions
    - d) Random chunks
- 56. Pages in paging are:
    - a) Variable sized
    - b) Fixed sized (typically 4KB)
    - c) Always 1MB
    - d) Determined by the application
- 57. The dirty bit in a page table entry indicates:
    - a) The page is in physical memory
    - b) The page has been modified
    - c) The page is readable
    - d) The page belongs to kernel
- 58. QEMU operates at which ring in KVM-based systems?
    - a) Ring 0 in root mode
    - b) Ring 3 in root mode
    - c) Ring 0 in VMX mode
    - d) Ring 3 in VMX mode
- 59. Xen is an example of:
    - a) Full virtualization
    - b) Paravirtualization

    - c) Container runtime
    - d) Operating system
- 60. Demand paging means:
    - a) All pages loaded at startup
    - b) Pages are created/loaded only when accessed
    - c) Pages are never swapped
    - d) Memory is unlimited
- 61. Movie-Star wants to run latest OS on VMs but has no hardware-assisted virtualization. Best option?
    - a) Hardware-assisted virtualization
    - b) Full virtualization (binary translation handles unsupported instructions)
    - c) Containers only
    - d) Bare metal deployment
- 62. Three major drawbacks of bare-metal servers vs hypervisors include all EXCEPT:
    - a) Inefficient resource utilization
    - b) Lack of flexibility and scalability
    - c) Better disaster recovery (bare metal has WORSE disaster recovery)
    - d) Increased infrastructure costs
- 63. IDT stands for:
    - a) Internet Data Transfer
    - b) Interrupt Descriptor Table
    - c) Input Device Table

- d) Internal Data Type

- 64. Which of the following causes a trap from user mode to kernel mode?
  - a) Reading a variable
  - b) Adding two numbers
  - c) Executing a privileged instruction / program fault / I/O interrupt
  - d) Declaring a string
- 65. In I/O virtualization, Direct I/O means:
  - a) I/O is emulated by the hypervisor
  - b) A slice of the device is directly assigned to each VM
  - c) I/O is disabled
  - d) All VMs share one device

Chapter 3: Containers & Docker

- 66. In container virtualization, what is the fundamental difference between PID namespaces and user namespaces?
  - a) PID namespaces isolate process identifiers while user namespaces isolate user/group identifiers and privileges
  - b) PID namespaces are for Docker, while user namespaces are for LXC
  - c) PID namespaces work in kernel space, user namespaces in user space
  - d) No difference
- 67. Which Linux kernel feature is primarily responsible for filesystem isolation in containers?
  - a) cgroups
  - b) Namespaces

    - c) KVM
    - d) SELinux
- 68. Cgroups are used to:
    - a) Isolate process visibility
    - b) Set resource limits (CPU, memory, network) on processes
    - c) Encrypt data
    - d) Manage network routing
- 69. Each process belongs to how many cgroups per controller hierarchy?
    - a) Multiple
    - b) Exactly one
    - c) Zero
    - d) Unlimited
- 70. The 'freezer' cgroup controller is used to:
    - a) Limit memory
    - b) Suspend running processes
    - c) Monitor CPU
    - d) Set network priority
- 71. In Docker architecture, which component creates new containers and then exits?
    - a) dockerd
    - b) containerd
    - c) runc

    - d) containerd-shim
- 72. containerd-shim's main role is to:
    - a) Build images
    - b) Become parent process of container, keep it running, handle streams
    - c) Create network namespaces
    - d) Pull images from registry
- 73. Which Dockerfile instruction sets the default command that can be overridden?
    - a) ENTRYPOINT
    - b) CMD
    - c) RUN
    - d) FROM
- 74. Which Dockerfile instructions create new image layers?
    - a) CMD, EXPOSE, WORKDIR
    - b) RUN, COPY, ADD
    - c) ENTRYPOINT, USER, CMD
    - d) FROM only
- 75. Docker images are stored and transferred as:
    - a) Single binary files
    - b) Layers
    - c) ZIP archives
    - d) Database records
- 76. A container registry is used to:

    - a) Run containers
    - b) Store, share, and version container images
    - c) Build source code
    - d) Monitor container health
- 77. PID 1 in Linux is:
    - a) The first user process
    - b) The init process and root of all processes
    - c) The kernel
    - d) The shell
- 78. Network namespace in containers provides:
    - a) Shared IP addresses
    - b) Private set of IP addresses, routing table, and sockets
    - c) Single network interface
    - d) No network isolation
- 79. Virtual Ethernet Link (veth) is used to:
    - a) Encrypt network traffic
    - b) Connect container network namespace with parent namespace
    - c) Monitor bandwidth
    - d) Block external traffic
- 80. The cpuset cgroup controller is used to:
    - a) Limit total CPU usage

    - b) Assign specific CPU cores to processes
    - c) Monitor CPU temperature
    - d) Increase CPU speed
- 81. Containers share the host OS:
    - a) Hardware only
    - b) Kernel
    - c) User applications
    - d) Desktop environment
- 82. The /proc folder in Linux contains:
    - a) User documents
    - b) Information about CPU, kernel, and running processes
    - c) Application binaries
    - d) Network configuration
- 83. Docker Hub is an example of a:
    - a) Container runtime
    - b) Public container registry
    - c) Container orchestrator
    - d) Build tool
- 84. The mount namespace isolates:
    - a) CPU resources
    - b) File system mount points
    - c) Network interfaces

    - d) User accounts
- 85. Docker volumes provide:
    - a) Temporary storage only
    - b) Persistent storage for containers
    - c) Network storage
    - d) Encrypted storage

Chapters 4-5: Cloud Providers & Architecture

- 86. AWS EC2 is equivalent to which Azure service?
    - a) Azure Functions
    - b) Azure Virtual Machine
    - c) Azure Blob Storage
    - d) Azure Kubernetes Service
- 87. S3 in AWS stands for:
    - a) Simple Server Service
    - b) Simple Storage Service
    - c) Secure Socket Server
    - d) Standard Storage System
- 88. OpenStack's compute component is called:
    - a) Swift
    - b) Neutron
    - c) Nova

    - d) Cinder
- 89. Neutron in OpenStack handles:
    - a) Compute
    - b) Networking
    - c) Block storage
    - d) UI dashboard
- 90. SLA stands for:
    - a) Software License Agreement
    - b) Service Level Agreement
    - c) System Load Average
    - d) Secure Login Access
- 91. In the hotel booking system, the Saga pattern is used for:
    - a) User authentication
    - b) Managing distributed transactions across microservices
    - c) Storing images
    - d) Caching data
- 92. The Outbox pattern ensures:
    - a) Messages are encrypted
    - b) Messages are reliably sent alongside business transactions
    - c) Messages are compressed
    - d) Messages are never lost from memory

- 93. An API Gateway primarily provides:
  - a) Database access
  - b) Routing, authentication, rate limiting, and request transformation
  - c) File storage
  - d) CPU scaling
- 94. REST API GET method is used to:
  - a) Create a new resource
  - b) Read existing resource content
  - c) Delete a resource
  - d) Update a resource
- 95. PostgreSQL was chosen for the hotel reservation system because of:
  - a) Its speed with unstructured data
  - b) Consistency requirement of reservation data (ACID)
  - c) Free pricing
  - d) Social media features
- 96. NoSQL databases scale:
  - a) Vertically (scale-up)
  - b) Horizontally (scale-out)
  - c) Cannot be scaled
  - d) Only in private cloud
- 97. Redis is primarily used as a:
  - a) Relational database

    - b) In-memory cache
    - c) File system
    - d) Message broker only
- 98. ACID stands for:
    - a) Automated, Consistent, Integrated, Durable
    - b) Atomicity, Consistency, Isolation, Durability
    - c) Available, Consistent, Independent, Distributed
    - d) Aggregated, Computed, Indexed, Dropped
- 99. Which AWS service provides serverless computing?
    - a) EC2
    - b) EBS
    - c) Lambda
    - d) S3
- 100. Google's equivalent of AWS EKS (Kubernetes) is:
    - a) GCE
    - b) GKE (Google Kubernetes Engine)
    - c) Cloud Functions
    - d) Cloud Storage

Chapter 6: Deployment & Kubernetes

- 101. What is the fundamental difference between Helm and Kustomize?
    - a) Helm uses YAML patches, Kustomize uses Go templates

    - b) Helm relies on templating with values.yaml, Kustomize uses declarative overlays
    - c) Helm requires Tiller server, Kustomize is client-side only
    - d) Kustomize supports rollbacks, Helm does not
- 102. When would you prefer Sagas over Two-Phase Commit (2PC) in microservices?
    - a) When transactions must be atomic across all services
    - b) When you need to avoid blocking locks and tolerate eventual consistency
    - c) When all services share the same database
    - d) When strong isolation is required
- 103. etcd in Kubernetes is:
    - a) A container runtime
    - b) A distributed key-value store for cluster data
    - c) A load balancer
    - d) A logging tool
- 104. Which Kubernetes component assigns pods to nodes?
    - a) kubelet
    - b) kube-proxy
    - c) kube-scheduler
    - d) kube-controller-manager
- 105. The smallest deployable unit in Kubernetes is:
    - a) Container
    - b) Pod

    - c) Node
    - d) Service
- 106. A Kubernetes DaemonSet ensures:
    - a) All pods run on one node
    - b) Exactly one pod runs on every (or selected) node
    - c) Pods have stable identities
    - d) Pods are never restarted
- 107. The Sidecar pattern in Kubernetes:
    - a) Runs before the main container
    - b) Runs alongside the main container in the same Pod
    - c) Replaces the main container
    - d) Runs on a separate node
- 108. Init containers must:
    - a) Run forever
    - b) Complete before app containers start
    - c) Run in parallel with app containers
    - d) Be written in Go
- 109. A Kubernetes Service of type NodePort:
    - a) Only allows internal cluster communication
    - b) Exposes the app on a port on each node
    - c) Creates a cloud provider load balancer
    - d) Disables external access

- 110. Canary deployment pattern:
  - a) Stops all old instances before deploying
  - b) Maintains two identical environments
  - c) Gradually releases new version to a small subset of users
  - d) Never updates existing servers
- 111. Blue/Green deployment requires:
  - a) No extra infrastructure
  - b) Double infrastructure (two environments)
  - c) Only one server
  - d) Manual file copying
- 112. Rolling deployment:
  - a) Causes full downtime
  - b) Replaces old version incrementally, no downtime
  - c) Uses two identical environments
  - d) Sends duplicate traffic
- 113. Immutable deployment:
  - a) Updates existing servers
  - b) Never updates existing servers, spins up new ones
  - c) Uses in-place upgrades
  - d) Only works with bare metal
- 114. Infrastructure as Code is NOT a benefit of:

    - a) Automation
    - b) Consistency
    - c) Manual configuration (IaC eliminates manual config)
    - d) Faster deployments
- 115. Terraform is:
    - a) AWS-only IaC tool
    - b) Multi-cloud IaC tool
    - c) A container runtime
    - d) A database
- 116. kubelet runs on:
    - a) Only the master node
    - b) Each worker node
    - c) Only in cloud
    - d) Only during deployment
- 117. kube-proxy maintains:
    - a) Container images
    - b) Network rules on nodes for pod communication
    - c) Database connections
    - d) User authentication
- 118. The Container Runtime Interface (CRI) is used to:
    - a) Build container images
    - b) Interface container runtime with kubelet

    - c) Encrypt containers
    - d) Monitor CPU usage
- 119. A StatefulSet provides:
    - a) Random pod names
    - b) Stable, unique identity and dedicated PersistentVolume per pod
    - c) No persistent storage
    - d) Only CPU scaling
- 120. Shadow deployment:
    - a) Replaces old version immediately
    - b) Sends duplicate traffic to new version but ignores responses
    - c) Only tests in staging
    - d) Requires zero infrastructure
- 121. The Adapter pattern in Kubernetes is used for:
    - a) Running init tasks
    - b) Transforming application output to standard format (logs, metrics)
    - c) Scaling pods
    - d) Authentication
- 122. Kubernetes follows a:
    - a) Imperative model
    - b) Declarative model (tell what you want, not how)
    - c) Manual model

    - d) Random model
- 123. Kubernetes self-healing means:
    - a) It fixes hardware failures
    - b) Automatically replaces failed containers and rolls back bad updates
    - c) It patches OS vulnerabilities
    - d) It writes new code
- 124. A service streaming logs from an app to a central system inside a K8s pod uses which pattern?
    - a) Init container
    - b) Sidecar pattern
    - c) Adapter pattern
    - d) StatefulSet
- 125. Testing new services by directing small traffic, increasing over time uses which deployment pattern?
    - a) Blue/Green
    - b) Recreate
    - c) Canary
    - d) Immutable

Chapter 7: Data Processing

- 126. A MapReduce job with 100 mappers and 10 reducers, each mapper emits 1 million pairs. How many per reducer (uniform distribution)?
    - a) 1 million
    - b) 10 million $(100 \times 1\text{M} = 100\text{M} \div 10 = 10\text{M})$

    - c) 100 million
    - d) 1 billion
- 127. In YARN, which component is responsible for negotiating resources and monitoring job execution?
    - a) NodeManager
    - b) Resource Manager
    - c) ApplicationMaster
    - d) JobHistoryServer
- 128. The default HDFS block size is:
    - a) 4KB
    - b) 1MB
    - c) 64MB or 128MB
    - d) 1GB
- 129. The default HDFS replication factor is:
    - a) 1
    - b) 2
    - c) 3
    - d) 5
- 130. The NameNode in HDFS:
    - a) Stores actual data blocks
    - b) Manages file metadata, creation, deletion, replication
    - c) Runs MapReduce jobs

    - d) Caches network packets
- 131. DataNodes send periodic:
    - a) Emails
    - b) Heartbeats to the NameNode
    - c) Logs to clients
    - d) Backups to secondary nodes
- 132. HDFS philosophy is:
    - a) Moving data is cheaper than moving computation
    - b) Moving computation is cheaper than moving data
    - c) All data should be centralized
    - d) Random access is preferred
- 133. The three phases of MapReduce are:
    - a) Input, Process, Output
    - b) Map, Shuffle & Sort, Reduce
    - c) Read, Write, Delete
    - d) Scan, Filter, Join
- 134. In the Map phase, each input pair produces:
    - a) Exactly one output pair
    - b) A list of intermediate key-value pairs
    - c) No output
    - d) A single sum

- 135. The Shuffle & Sort phase:
  - a) Compresses data
  - b) Groups intermediate pairs by key
  - c) Deletes duplicate keys
  - d) Encrypts data
- 136. The Reduce phase:
  - a) Splits data into smaller parts
  - b) Combines grouped values into final output
  - c) Maps input to intermediate pairs
  - d) Sorts alphabetically
- 137. Secondary NameNode in HDFS:
  - a) Replaces the NameNode immediately on failure
  - b) Periodically merges EditLog and FsImage to prevent EditLog from growing endlessly
  - c) Stores user data
  - d) Runs MapReduce jobs
- 138. Cross-rack communication in HDFS is:
  - a) Free
  - b) More expensive than within-rack communication
  - c) Not possible
  - d) Faster than within-rack
- 139. YARN's Resource Manager contains which two components?

    - a) NameNode and DataNode
    - b) Scheduler and Application Manager
    - c) Container and Pod
    - d) Master and Worker
- 140. CLC in YARN stands for:
    - a) Container Load Configuration
    - b) Container Launch Context
    - c) Central Log Collector
    - d) Cluster Level Command
- 141. Batch processing is best described as:
    - a) Real-time event handling
    - b) Processing large amounts of data at once to deliver results
    - c) Processing one record at a time
    - d) Interactive querying
- 142. Stream processing is used for:
    - a) Generating monthly reports
    - b) Real-time dashboards, alerts, anomaly detection
    - c) Archive storage
    - d) Backup operations
- 143. Apache Flink's advantage over Spark is:
    - a) Better batch processing
    - b) True streaming (not micro-batches)

    - c) Simpler API
    - d) Lower cost
- 144. HDFS data integrity is maintained by:
    - a) Encryption
    - b) Calculating checksum within each replica
    - c) User verification
    - d) Manual inspection
- 145. Structured data example:
    - a) Raw videos
    - b) Transaction data in tables
    - c) Social media posts
    - d) Sensor readings

Chapter 8: Cloud Security

- 146. In the Authorization Code flow, what is the role of the authorization code?
    - a) Used to authenticate the resource owner
    - b) It is exchanged for an access token
    - c) It acts as the access token
    - d) It contains the user's credentials
- 147. OAuth 2.0 client credentials flow's client_secret is compromised. Immediate impact?
    - a) All user sessions become invalid
    - b) Attacker gains API access as the client

    - c) The authorization server becomes compromised
    - d) Only cached tokens are affected
- 148. What does GDPR primarily regulate in cloud?
    - a) Network bandwidth
    - b) Personal data processing and privacy
    - c) Cloud pricing models
    - d) Software licensing
- 149. Which regulation primarily affects healthcare organizations in the US?
    - a) PCI DSS
    - b) HIPAA
    - c) SOX
    - d) FISMA
- 150. Symmetric encryption uses:
    - a) Public-private key pair
    - b) Same key for encrypt and decrypt
    - c) No keys
    - d) Three different keys
- 151. RSA is an example of:
    - a) Symmetric encryption
    - b) Asymmetric encryption
    - c) Hashing algorithm

    - d) Compression
- 152. AES-256 is commonly used for:
    - a) Data in transit
    - b) Data at rest (encryption)
    - c) Network routing
    - d) Load balancing
- 153. TLS/SSL creates:
    - a) Encrypted storage
    - b) Secure communication channel between two nodes
    - c) Database connections
    - d) User accounts
- 154. Digital signature proves:
    - a) Message came from specific sender, was not changed, cannot be denied
    - b) Message was encrypted
    - c) Message was compressed
    - d) Message was cached
- 155. A root certificate is:
    - a) Signed by an intermediate CA
    - b) Self-signed and pre-trusted by browsers/OS
    - c) Signed by the website owner
    - d) Never used
- 156. MFA combines:

    - a) Two databases
    - b) Two or more authentication methods
    - c) Two servers
    - d) Two networks
- 157. Session-based authentication stores sessions on:
    - a) Client side only
    - b) Server side
    - c) Neither
    - d) In DNS
- 158. JWT is considered stateless because:
    - a) It requires server-side storage
    - b) No session info stored on server; token is self-contained
    - c) It uses cookies
    - d) It requires a database
- 159. JWT payload "iss" claim stands for:
    - a) Issue date
    - b) Issuer
    - c) Is secure
    - d) Instance
- 160. JWT "exp" claim stands for:
    - a) Experience

    - b) Expiration time
    - c) Export
    - d) Exception
- 161. RBAC stands for:
    - a) Rule-Based Access Control
    - b) Role-Based Access Control
    - c) Resource-Based Access Control
    - d) Registry-Based Access Control
- 162. DAC allows:
    - a) System admin to manage all access
    - b) Resource owners to grant/revoke access to their own files
    - c) Automatic access for everyone
    - d) No access control
- 163. OAuth 2.0's primary purpose is:
    - a) Authentication
    - b) Authorization (letting apps access resources without sharing passwords)
    - c) Encryption
    - d) Logging
- 164. The Client Credentials grant type is used for:
    - a) User login with Google
    - b) Machine-to-machine communication (no user involved)
    - c) Frontend-only apps

    - d) Mobile apps
- 165. The Implicit Grant type is:
    - a) The most secure OAuth2 flow
    - b) Deprecated due to security concerns
    - c) Used for server-to-server
    - d) The default flow
- 166. OIDC differs from OAuth2 by adding:
    - a) Encryption
    - b) Identity layer (ID Token)
    - c) Better performance
    - d) Database access
- 167. SSO stands for:
    - a) Simple Server Operation
    - b) Single Sign-On
    - c) Secure Socket Operation
    - d) System Security Option
- 168. GDPR breach notification must be made within:
    - a) 24 hours
    - b) 72 hours
    - c) 1 week
    - d) 30 days

- 169. GDPR maximum penalty for major violations:
  - a) €1 million
  - b) €20 million or 4% of global annual revenue
  - c) €100,000
  - d) €5 million
- 170. PCI DSS applies to organizations that:
  - a) Process healthcare data
  - b) Store, process, or transmit credit card data
  - c) Operate social media
  - d) Sell cloud services
- 171. PCI DSS Level 1 requires:
  - a) Self-assessment
  - b) Onsite assessment by QSA for over 6 million transactions/year
  - c) No assessment
  - d) Online quiz
- 172. JWT signature is created using:
  - a) Encoded header + encoded payload + private key
  - b) Just the payload
  - c) Just the header
  - d) The user's password
- 173. Federated Identity means:
  - a) Each system has its own login

    - b) Single digital identity across multiple systems/organizations
    - c) No authentication needed
    - d) Passwords are shared between systems
- 174. Input validation and sanitization prevents:
    - a) Network latency
    - b) SQL injection and other injection attacks
    - c) Hardware failure
    - d) Data duplication
- 175. Secrets management tools include:
    - a) HashiCorp Vault, AWS Secrets Manager
    - b) Microsoft Word, Notepad
    - c) Docker Hub
    - d) JIRA
- 176. In a CP system (CAP theorem), during a network partition:
    - a) System remains available but returns stale data
    - b) System rejects writes until partition heals
    - c) System prioritizes availability
    - d) System switches to CA mode
- 177. For financial data with conflicting values after partition, the best resolution is:
    - a) Last-writer-wins
    - b) CRDTs

    - c) Manual intervention with audit trail
    - d) Average values
- 178. JWT's "sub" claim refers to:
    - a) Subject/User ID
    - b) Subscription
    - c) Submit date
    - d) Sub-domain
- 179. Access Token vs Refresh Token:
    - a) Access token is long-lived, refresh is short-lived
    - b) Access token is short-lived for APIs, refresh token is long-lived to get new access tokens
    - c) Both are identical
    - d) Refresh tokens are deprecated
- 180. GDPR principle of "Data Minimization" means:
    - a) Store as much data as possible
    - b) Only collect data that is necessary for the intended purpose
    - c) Minimize database size
    - d) Delete all data after 24 hours
- 181. Asymmetric encryption is often used to:
    - a) Encrypt large datasets
    - b) Securely exchange symmetric keys
    - c) Replace symmetric encryption

    - d) Compress data
- 182. SAML 2.0 is used for:
    - a) Web API authorization
    - b) Enterprise SSO, XML-based
    - c) Mobile authentication
    - d) Password management
- 183. Physical security threats to data centers include all EXCEPT:
    - a) Unauthorized access
    - b) Natural disasters
    - c) SQL injection (this is a software threat, not physical)
    - d) Hardware theft
- 184. The Shared Responsibility Model involves:
    - a) Cloud Provider, Cloud Developers, Cloud Consumers
    - b) Only the cloud provider
    - c) Only the user
    - d) Government only
- 185. Certificate Authority (CA) verifies:
    - a) IP addresses
    - b) Identity of the certificate requester and issues certificates
    - c) Database integrity
    - d) Network speed

Mixed Application Questions

- 186. A cloud orchestration system with Region A (low latency, high cost), Region B (moderate), Region C (high latency, low cost). For latency-sensitive variable load app, which optimizes cost while maintaining SLA?
    - a) Always use Region A
    - b) Use Region A for peak, Region B for normal, Region C for batch
    - c) Distribute equally
    - d) Use Region C primarily, burst to Region A
- 187. A cron job calling a third-party API (backend-to-backend) should use which OAuth2 grant type?
    - a) Authorization Code
    - b) Implicit
    - c) Client Credentials
    - d) Device Code
- 188. A user saving schedules to Google Calendar (on behalf of user, already registered) uses which OAuth2 grant?
    - a) Client Credentials
    - b) Authorization Code
    - c) Implicit
    - d) Resource Owner Password
- 189. For TechFlow Solutions project management tool, which access control model is most suitable?
    - a) DAC
    - b) RBAC (Role-Based - enterprise with different roles like manager, developer)

    - c) ABAC
    - d) No access control needed
- 190. Authorization Code Grant differs from Implicit by:
    - a) Having an extra step (exchanging code for token via backend)
    - b) Being less secure
    - c) Not requiring user login
    - d) Using XML
- 191. A service inside a K8s pod needs to stream logs to a central system. Best design pattern?
    - a) Init Container
    - b) Sidecar Pattern
    - c) StatefulSet
    - d) DaemonSet
- 192. The distributed key-value store Kubernetes uses as its primary backing store is:
    - a) Redis
    - b) MongoDB
    - c) etcd
    - d) Cassandra
- 193. For designing a cloud-based system, the advantage of cloud over traditional methods includes all EXCEPT:
    - a) Scalability
    - b) Lower initial cost

    - c) Global accessibility
    - d) No internet dependency (cloud REQUIRES internet)
- 194. In a microservices architecture, each service should have:
    - a) Shared database
    - b) Its own database (Database per service pattern)
    - c) No database
    - d) Direct hardware access
- 195. The three types of data in Big Data (the 3 Vs) are:
    - a) Volume, Velocity, Variety
    - b) Value, Vision, Validation
    - c) Verbose, Virtual, Volatile
    - d) Version, Vendor, Visibility
- 196. Table Q2f comparison: Type 1 hypervisor Performance, Startup, Overhead, Isolation?
    - a) High, Fast, Low, Strong
    - b) High performance, Slow startup, Moderate overhead, Strong isolation
    - c) Moderate, Fast, High, Low
    - d) Low, Slow, High, Low
- 197. Container Performance, Startup, Overhead, Isolation?
    - a) High performance, Fast startup, Low overhead, Low isolation
    - b) Low, Slow, High, Strong
    - c) Moderate, Moderate, Moderate, Moderate
    - d) High, Fast, Low, Strong

- 198. DAG stands for (used in Spark/Flink):
  - a) Data Access Gateway
  - b) Directed Acyclic Graph
  - c) Distributed Application Grid
  - d) Dynamic Allocation Group
- 199. Kubernetes ClusterIP service type provides:
  - a) External access via node ports
  - b) Internal communication within the cluster only
  - c) Cloud load balancer integration
  - d) DNS resolution only
- 200. PCI DSS has how many core requirements grouped into how many goals?
  - a) 10 requirements, 4 goals
  - b) 12 requirements, 6 goals
  - c) 8 requirements, 3 goals
  - d) 15 requirements, 5 goals